Final Project: UGV

Project Brief for All Teams

1. Overview

For the final project, every group will design, build, and test a **UGV (Unmanned Ground Vehicle)**: a ground robot with an onboard camera that can sense and act on its own.

Deadline: Friday, October 23.
Competition: Saturday, October 24, at the college.

2. Group Structure

Each group will consist of four teams:

- **Control:** programming, communication, and the vehicle's control logic.
- **Electrical:** wiring, power, motor drivers, and connecting the electronics.
- **Mechanical:** the chassis, wheels/tracks, motor mounting, and the overall build.
- **Computer Vision:** processing the ESP32-CAM video feed (e.g. detection and tracking).

If a member is part of more than one subteam, they can also work on the other subteam's tasks, but **only after finishing the tasks of their main subteam**.

3. Project Requirements

The project **must include**:

- **ESP32**
- **ESP32-CAM** (camera)

The overall weight of the UGV **must not exceed 2 kg**.

4. Mission

The UGV must detect a **colored flag** and **announce which color it is**. It must then use a **sensor** to **track the flag** and **stay 50 cm away from it**.

5. Scoring

Category	Points
Control	10
Electrical	10
Mechanical	10
Computer Vision	10
Documentation	10
Total	**50**
Bonus: creative ideas	Bonus points

6. What We Expect From Each Group

- Build the vehicle and assemble the electronics.
- Program the ESP32 and ESP32-CAM so the UGV can detect the flag, announce its color, and keep 50 cm from it.
- Test and improve the vehicle until it works reliably.
- **Document every step** of the project: design decisions, wiring, code versions, test results, problems you faced, and how you fixed them (notes, photos, and screenshots are welcome).

7. Groups

Group	Control	Electrical	Mechanical	Computer Vision
Group 1	Youssef Sherif		Mohamed Ossama Bassant	Shrouq
Group 2	Saysban	Yamen	Ziad Roushdy Salem Ebeid	Gedawy
Group 3	Ali Fawzy	Ahd Atia	Rahma Jana	Tasneem Habiba
Group 4	Youssef Gomaa	Malak Ahmed	Karim Farah	Zeina Rabbany
Group 5	Abdelrahman Ashraf Asma Raafat	Omar Mohamed	Wagdy Nabil	Omar Metwally
Group 6	Jasmine Yaseen Moataz	Rodina	Yaseen Aly	Fares Hazem Khaled
Group 7	Ahmed Shahwan	—	Ahraquat Mariam Aly	Nour Amr Asmaa

8. Contacts

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